

#Aysha Bilai #100916114

#####Q1#####

```
#user inputs a positive integer
integer = int(input("Enter a positive integer: "))
```

```
#determine if user input is prime or composite
def is_prime(num):
```

```
    if num < 2:
        return False
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            return False
    return True
```

```
if integer <= 0:
    print("Enter a positive integer.")
elif is_prime(integer):
    print(f'{integer} is a prime number.')
else:
    print(f'{integer} is a composite number.')
```

#####end#####

#####Q2#####

```
# User input for a positive integer n
even_numbers = [] # Initialize the list for even numbers
```

```
while True:
    user_input = input("Enter a positive integer (type 'exit' to quit): ")
```

```
    if user_input.lower() == 'exit': # Exits loop if 'exit' is inputted
        break
```

```
    try:
        number = int(user_input)
        if number > 0:
            if number % 2 == 0: # Isolate the even numbers
                even_numbers.append(number)
        else:
            print("Please enter a positive integer.")
    except ValueError:
```

```

        print("Invalid input. Please enter a valid integer.")

sum_of_even_numbers = sum(even_numbers)
print("Entered even numbers:", even_numbers)
print("Sum of even numbers:", sum_of_even_numbers)

#####end#####

#####Q3#####

print("Can you guess which number (1-100) I am thinking of?")

# import random library
import random

# python chooses a random number
rand_num = random.randint(1, 100)

# user begins guessing
while True:
    user_guess = int(input("Enter your guess: "))

    # check if the guess is correct
    if user_guess == rand_num:
        print("Congratulations! You guessed the correct number.")
        break
    elif user_guess < rand_num:
        print("Too low. Try again.")
    else:
        print("Too high. Try again.")

#####end#####

#####Q4#####

print("You are player A")
print("First to roll a six on the six-sided die wins.")
time.sleep(1.5)

# import random library
import random

#implement time delay
import time

```

```
while True:
    # Player A rolls the dice
    A_rolls = random.randint(1, 6)
    print("Player A rolled...", A_rolls)
    time.sleep(2)

    # Check if Player A rolled a six
    if A_rolls == 6:
        print("Congratulations, Player A! You win!")
        break
    else:
        print("Player B's turn.")
        time.sleep(1)

    # Player B rolls the dice
    B_rolls = random.randint(1, 6)
    print("Player B rolled...", B_rolls)
    time.sleep(2)

    # Check if Player B rolled a six
    if B_rolls == 6:
        print("Congratulations, Player B! You win!")
        break
    else:
        print("Player A's turn.")
        time.sleep(1)

#####end#####
```